4 wireless physical layer

Problem 4.1 : a wifi access point transmits @ $P_t = 20$ dBm with antenna gain $G_t = 2$ dBi. a laptop @ $d = 50$m has antenna gain $G_r = 0$ dBi. carrier frequency $f = 2.4$ GHz

a) compute free space path loss L_{FSPL} in dB

GHz → MHz m → km
2.4 → 2400 50 → 0.05

$L_{FSPL} = 32.43 + 20\log_{10}(2400) + 20\log_{10}(0.05) = 74.03$

b) compute received power P_r in dBm

$P_r = P_t + G_t + G_r - L_{FSPL}$

$P_r = 20 + 2 + 0 - 74 = -52$

c) minimum sensitivity of laptop receiver is 80 dBm. link margin?

$-80 - (-52) = -28 \longrightarrow 28$ dB headroom

d) AP → $d = 200$ m @ same frequency, how many dB does L_{FSPL} increase

path loss increased by $20\log\left(\frac{200}{50}\right) = 12$ dB increase

Problem 4.2

a) how many bits per symbol does QPSK carry? 16 QAM? 64 QAM?

BPSK → 1 bit/sym 16-QAM → 4 bit/sym
QPSK → 2 bit/sym 64-QAM → 6 bit/sym

b) a 64 QAM symbol is transmitted @ 10 M symbols/s. (symbol) what is the bit rate?

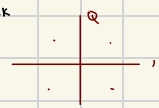
bit rate = symbol rate × bits per symbol

$= 10\,\frac{M}{s} \times 6\,\frac{b}{M} = 60\ b/s$

c) wifi uses adaptive modulation. when a user moves far from access point, the system switches from 256 QAM to BSPK. why?

less bits per symbol → farther

d) sketch the 4 symbol constellation for QPSK



Problem 4.3 : a cellular network uses a reuse factor of $N = 7$ and a cell radius $R = 500$ m. Path-loss exponent is $n = 3.5$

a) compute reuse distance

$D = R\sqrt{3N} = 500\sqrt{21} = 2291$ m

b) estimate the carrier to interference ratio (C/I) using the approximation $C/I = \frac{\left(\frac{3}{N}\right)^{n/2}}{6}$

$\frac{C}{I} \approx \frac{(3N)^{n/2}}{6} = \frac{(3 \cdot 7)^{\frac{3.5}{2}}}{6} = 34.3$

c) if operator switches to $N = 3$, qualitatively what happens to
↳ 7 → 3

(i) capacity per unit area

higher capacity because less sharing

↑N → ↓ reuse, ↓ capacity

↓N → ↑ reuse, ↑ capacity

(ii) interference

more interference